AI Critique of the Use Case Diagram

Prepared per Lab 1 Manual, Section 3.5 (Get an AI critique of your Use Case diagram)

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1. Method

To get an independent read, the critique below was produced by a fresh AI reviewer instance that was given **only** two attached files — the use case diagram image (`use_case_diagram.png`) and the functional/non-functional requirements document (`requirements_draft.md`) — and no other context about how the diagram was designed or built. This mirrors uploading the two files to a fresh AI chat session as instructed in the manual.

2. Prompt Given to the AI

> *"Critique the attached use case diagram image, with respect to how well it matches the attached requirements file."*

3. AI's Response (verbatim)

> Overall the diagram captures the core gaze→scroll loop reasonably well, but there are several gaps and notation issues worth flagging against the requirements draft.
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> **Coverage gaps.** I don't see anything for FR-2 (disabling autoscroll and messaging the user when camera access is denied), FR-5/NFR-6 (prompting recalibration or pausing on low confidence / no face detected), FR-15 (re-running content classification after scroll/navigation), FR-21 (per-direction enable/disable), FR-25 (the status indicator showing active/paused/recalibrating), or NFR-11 (persisting configurable settings across sessions). "Configure Autoscroll Settings" is floating with no include/extend to the parameters it's supposed to cover (sensitivity, speed, per-direction toggles), so it reads as a stub rather than a modeled use case. Given how much the requirements emphasize graceful degradation (NFR-6) and low-confidence handling (FR-5, FR-11), the absence of an explicit "Handle Low Confidence / Pause on Tracking Loss" use case is the biggest hole.
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> **"Recalibrate Gaze Tracking" extending "Estimate Gaze Point" is backwards.** Per FR-5, recalibration is triggered by the system when confidence drops during gaze estimation — that's the base flow needing recalibration as an exceptional branch, so the extend arrow should point from Recalibrate Gaze Tracking toward Estimate Gaze Point conceptually, but more importantly Recalibrate Gaze Tracking should itself include Calibrate Gaze Tracking (it's presumably re-running the same calibration sequence), and that relationship is missing entirely — right now it's a dead-end bubble.
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> **"Pause/Resume Autoscroll" extending "Estimate Gaze Point"** is questionable. FR-24 says pause/resume is a user-initiated control action, not a variant of gaze estimation; it should extend or

associate with **Autoscroll Page**, not the gaze-estimation use case, since pausing is about suspending the autoscroll behavior, not gaze sensing itself.

The abstract "Scroll Page" generalization is a good modeling choice for FR-17/FR-20 (both Autoscroll Page and Manual Scroll issue scroll commands), but Manual Scroll has no actor association drawn to User even though FR-20 explicitly makes manual scroll a user action that takes precedence over autoscroll — that precedence/cooldown relationship (FR-20's second clause) isn't represented anywhere, and Manual Scroll appears to only connect to Display/OS, which is odd since the user is the one initiating it.

Actor modeling: Webcam and Display/OS as actors are fine (external systems), but Calibrate Gaze Tracking associating directly with Webcam while skipping the "continuously capture video frames" (FR-6) step is a minor omission — there's no distinct "Capture Video Frame" use case, so FR-6 is implicitly folded into Estimate Gaze Point without being named.

Use cases with weak grounding: "Identify Important Content Regions" is well-grounded (FR-13/FR-14), but its «include» from Autoscroll Page should arguably also feed into a content-change trigger per FR-15, which isn't shown.

Happy to sketch a revised version incorporating these fixes if that would help.

4. Selected Aspect: Unhelpful / Factually Wrong

> ***"'Pause/Resume Autoscroll' extending 'Estimate Gaze Point' is questionable. FR-24 says pause/resume is a user-initiated control action, not a variant of gaze estimation; it should extend or associate with Autoscroll Page, not the gaze-estimation use case, since pausing is about suspending the autoscroll behavior, not gaze sensing itself."***

5. Justification

We selected this point because it is a clear, checkable misread rather than a matter of opinion. In our actual diagram, `Pause / Resume Autoscroll` (UC-9) does «extend» `Autoscroll Page` (UC-7) — exactly the relationship the AI recommends — not `Estimate Gaze Point` (UC-4). The AI's misreading traces to a real legibility problem we can point to directly: the UC-9→UC-7 extend line is drawn as a straight vertical line that passes visually behind the "Capture Screen Content" oval, in the same crowded region where the genuine UC-10→UC-4 (Recalibrate Gaze Tracking «extend» Estimate Gaze Point) relationship is drawn and labelled. With four ellipses and two crossing dashed lines packed into a small area, it is easy to see how an automated (or human) reader could attribute the wrong endpoint to the wrong line.

This is useful precisely because it is wrong about the specific relationship: it tells us the diagram's layout in that bottom cluster is dense enough to cause misattribution, which is a legitimate diagram-quality issue distinct from the modelling decision itself. We plan to widen the spacing around Estimate Gaze Point, Pause/Resume Autoscroll and Recalibrate Gaze Tracking when the Use Case model is elaborated in Lab 2, even though the underlying relationship the AI proposed already matches

our design.

Note: per the Lab 1 manual, the use case diagram and requirements files referenced above are submitted separately as deliverables and are not reproduced in full in this report.